

Universal architecture of corneal segmental tomography biomarkers for artificial intelligence-driven diagnosis of early keratoconus

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ABSTRACT

Aims To develop a comprehensive three-dimensional analyses of segmental tomography (placido and optical coherence tomography) using artificial intelligence (AI).

Methods Preoperative imaging data (MS-39, CSO, Italy) of refractive surgery patients with stable outcomes and diagnosed with asymmetric or bilateral keratoconus (KC) were used. The curvature, wavefront aberrations and thickness distributions were analysed with Zernike polynomials (ZP) and a random forest (RF) AI model.

For training and cross-validation, there were groups of healthy (n=527), very asymmetric ectasia (VAE; n=144) and KC (n=454). The VAE eyes were the fellow eyes of KC patients but no further manual segregation of these eyes into subclinical or forme-fruste was performed.

Results The AI achieved an excellent area under the curve (0.994), accuracy (95.6%), recall (98.5%) and precision (92.7%) for the healthy eyes. For the KC eyes, the same were 0.997, 99.1%, 98.7% and 99.1%, respectively. For the VAE eyes, the same were 0.976, 95.5%, 71.5% and 91.2%, respectively. Interestingly, the AI reclassified 36 (subclinical) of the VAE eyes as healthy though these eyes were distinct from healthy eyes. Most of the remaining VAE (n=104; forme fruste) eyes retained their classification, and were distinct from both KC and healthy eyes. Further, the posterior surface features were not among the highest ranked variables by the AI model.

Conclusions A universal architecture of combining segmental tomography with ZP and AI was developed. It achieved an excellent classification of healthy and KC eyes. The AI efficiently classified the VAE eyes as 'subclinical' and 'forme-fruste'.

INTRODUCTION

High-resolution optical coherence tomography (OCT) imaging of the human cornea has provided unprecedented access to the corneal layers for clinical decision making.¹ Each of these layers undergo a distinct remodelling with the onset of a disease and after surgery.^{1,2} Keratoconus (KC) is one such corneal degenerative condition which has a profound effect on the thickness and shape of corneal layers. Hence, OCT tomography of the cornea can be regarded as 'segmental layered tomography' as layers of the cornea can be evaluated separately and also with good repeatability.^{3,4} Artificial intelligence (AI) and deep learning (DL) have gained much traction for detection of early KC using either topography alone or tomography.⁵

However, a universal architecture of segmental layered tomography variables for inclusion in AI/DL models is still lacking since different devices use different indices for monitoring progression and diagnosis of KC. Further, the disease progression may have a differential impact on each corneal layer, and may apply to any one or a combination of curvatures, thicknesses and wavefront aberrations. Furthermore, none of the AI/DL models incorporate the complete three-dimensional (3D) spatial variation in curvatures and thicknesses of the corneal layers. It should be noted that the elevation map of the posterior corneal surface provided by commercial tomographers may not include a 3D axial offset to factor corneal thickness distribution. Thus, only the shape of the posterior corneal surface can be analysed and not the 3D thickness distribution. Therefore, vital tomographic features of early KC may be missed.

We recently showed that segmental tomography of the epithelium and stroma could be useful in detecting early KC changes with high accuracy.^{3,6} We also showed that 3D mapping of the epithelium thickness was sensitive to specific changes after different types of refractive surgery.^{7,8} The aim of this study was to develop a universal architecture of segmental layered tomography variables and test them on a large population of healthy, asymmetric fellow eyes of KC patients and clinically diagnosed KC corneas. This is the first study of its kind where 3D distribution of curvatures and aberrations of different surfaces, thicknesses and volumes of epithelium and stroma were combined. Clinical measurements from a hybrid tomographer (MS-39; CSO, Florence, Italy) and a unique combination of Zernike polynomials were used for all the analyses.

MATERIALS AND METHODS

This study was a retrospective review of hospital records. The preoperative tomography of healthy eyes (n=527 patients with one eye per patient), which underwent refractive surgery previously and had at least 1 year of follow-up data with stable refractive outcomes, were included. KC patients were included in the study. A total of 310 KC patients had the disease in both eyes and one eye from each patient was included at random (coin toss). A total of 144 KC patients had clinically diagnosed KC in one eye. The other eye had a corrected distance visual acuity of 6/6, appeared either healthy (subclinical) on slit lamp and tomography



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(corneal thickness, anterior and posterior surface curvature) or had signs of early disease (forme fruste) on visual examination of tomography by an experienced surgeon (RS). Both eyes of these patients were included in this study. However, the fellow eye was classified as very asymmetric ectasia (VAE) only and the other eye was included as KC.³ Thus, the total number of KC eyes included in this study was 454 (=310+144). A recent review revealed significant overlap between the definitions of sub-clinical disease and forme fruste KC (FFKC) in published literature.⁹ For example, ~22% and ~32% of the published studies on subclinical disease and FFKC used a normal slit lamp examination of the cornea as a decision criterion.⁹ Some studies compared healthy with FFKC and healthy with KC eyes but did not compare FFKC with KC eyes to assess the overlap of diagnostic variables between FFKC and KC.¹⁰ In our study, the VAE group was considered as one group of eyes. The exclusion criteria included any prior corneal surgery, ocular surgery, autoimmune disease, pregnancy and any active ocular inflammation at the time of or 3 months prior to imaging.

The study exclusively used data from a hybrid tomographer (MS-39) since it measured the anterior surface curvature with Placido technique while the thicknesses of layers (epithelium and stroma) and posterior surface curvature were measured with high resolution OCT. A maximum of 31232 points were evaluated on the anterior surface by MS-39. The high-resolution OCT used a centre wavelength of 845 nm and achieved an axial optical resolution of 3.6 µm in tissue. A total of 12 radial B-scans (24 hemimeridians) and a maximum scan diameter of 10 mm were the OCT imaging settings. All B-scans were evaluated for missing edge detection.¹¹ We evaluated the following variables for a universal architecture since these variables were primarily derived from Zernike polynomials only:

- Curvature: From the Placido imaging data, the elevation (x, y and z coordinates) of anterior (Ant) corneal surface was derived by the MS-39 software. From the OCT B-scans, the epithelium-Bowman's interface was detected and the epithelium thickness along the direction of the normal to the anterior corneal edge was computed by the MS-39 software.^{7 8} Similarly, the posterior corneal edge was detected and the

total corneal thickness along the direction of the normal to the anterior corneal edge was computed by the MS-39 software. Then, the Ant elevation, posterior corneal surface (post) elevation, epithelium thickness and total corneal thickness (=epithelium + stroma) data were exported from MS-39 as comma separated value files. It should be noted that curvature includes the first and second order spatial gradient of elevation of Ant, Bowman's layer (BW) and post. All the elevation data were resampled using Zernike polynomials, where the z co-ordinate was mapped as a function of x and y co-ordinates. The elevation of the epithelium-BW interface was obtained by adding the epithelium thickness to the Ant elevation along the direction of the local normal. Then, the curvature of Ant, BW and post were computed using the same polynomials.^{3 6} For the curvature of Ant and BW, an incident medium of air was assumed and the net corneal refractive index of 1.3375 was used.^{3 6} For the curvature of Post, a corneal refractive index of 1.376 and an aqueous refractive index of 1.336 was assumed.

- Ray tracing for wavefront: In addition to curvature, ray tracing was used to determine the wavefront aberrations of Ant, BW and total cornea (3D). Here, the total cornea aberrations included the post wavefront aberrations also since 3D ray tracing of the whole cornea was performed. For independent ray tracing of Ant and BW, a refractive index of 1.3375 was used.^{3 6} For total cornea, a refractive index of 1.401 and 1.376 was used for the epithelium and stroma, respectively. For the aqueous, a refractive index of 1.336 was assumed. The aberrations were further decomposed with Zernike polynomials. The root mean squares of lower and higher order aberrations were calculated, for example, AntLORMS, AntHORMS, BWLORMS, BWHORMS, 3D-LORMS, 3D-HORMS (table 1).
- 3D spatial mapping of thicknesses: In our recent work, we showed that the Zernike polynomials can be used for the spatial mapping of thickness distribution.^{7 8 12} The thickness distribution implies the spatial variation in thickness as a function of radius and meridian. The thickness of epithelium and cornea at a given location was calculated along

Table 1 A summary of all variables derived from segmental tomography and Zernike analyses

Anterior surface	Epithelium-BW interface	Posterior surface	Corneal thickness	Epithelium thickness	Volume
AntK1	BWK1	PostK1	CCT	CET	EpiVol
AntK2	BWK2	PostK2	Min CT	Min ET	CorVol
AntAxis	BWAxis	PostAxis	RatioCT	RatioET	Epi-Cor-Vol-Ratio
AntKmax	BWKmax	PostKmax	—	—	EpiVolRatio
AntLORMS*	BWLORMS†	3D-LORMS‡	tLORMS§	eLORMS¶	CorVolRatio
AntHORMS*	BWHORMS†	3D-HORMS‡	tHORMS§	eHORMS¶	—
AntComaRMS*	BWComaRMS†	3D-ComaRMS‡	tComaRMS§	eComaRMS¶	—
AntDefocus*	BWDefocus†	3D-Defocus‡	tDefocus§	eDefocus¶	—
AntSA*	BWSA†	3D-SA‡	tSA§	eSA¶	—

*Anterior surface wavefront aberrations.

†Epithelium-BW wavefront aberrations.

‡Total corneal wavefront aberrations including epithelium and stroma as separate layers. These aberrations include refraction due to both anterior and posterior corneal surfaces.

§Total corneal thickness distribution mapped by Zernike polynomials.

¶Epithelium thickness distribution mapped by Zernike polynomials.

Ant, anterior corneal surface; BW, Epithelium-Bowman's layer interface; CCT, central corneal thickness; CET, central epithelium thickness; ComaRMS, root mean square of coma Zernike terms ($Z_{-3,-1}^3$ and $Z_{+3,+1}^3$); CorVol, Volume of cornea (epithelium plus stroma); CorVolRatio, Ratio of volume of cornea in superior half to inferior half; 3D, Total cornea including epithelium and stroma; e, Epithelium thickness; Epi-Cor-Vol-Ratio, Ratio of Epithelium to Corneal volume; EpiVol, Volume of epithelium; EpiVolRatio, Ratio of volume of epithelium in superior half to inferior half; ET, epithelium thickness; HORMS, root mean square of higher order Zernike terms; K1, steep axis curvature; K2, flat axis curvature; LORMS, root mean square of lower order Zernike terms; min, minimum value of parameter; Post, posterior corneal surface; RatioCT, ratio of CCT to min CT; RatioET, ratio of CET to min ET; SA, spherical aberration (fourth order); t, total corneal thickness.

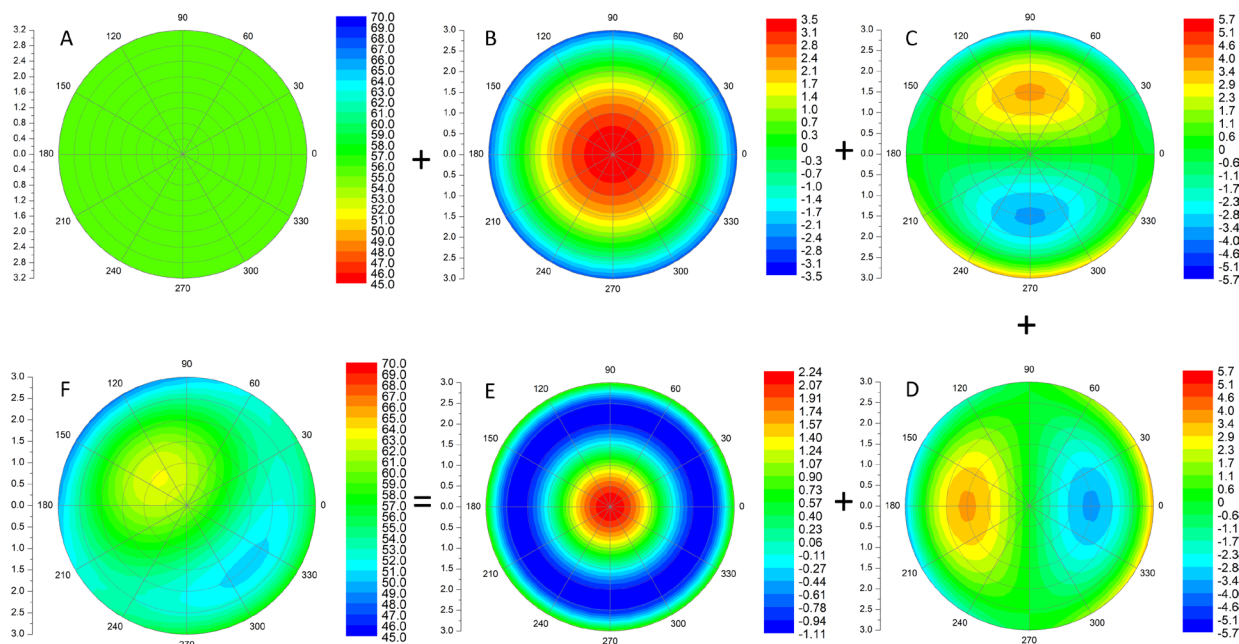


Figure 1 An example of Zernike decomposition of 3D spatial thickness distribution of the epithelium: (A) piston distribution; (B) defocus distribution; (C) vertical coma distribution; (D) horizontal coma distribution; (E) spherical aberration distribution; (F) net epithelium thickness distribution obtained by adding A–E. 3D, three dimensional.

the direction of the normal to the anterior corneal edge at the same location. Here, the Zernike decomposition can be written as:

$$t = \sum_{m=0}^M \sum_{n=-m}^{+m} a_n^m Z_n^m \quad (1)$$

Here, t was the thickness measured at a specific radius and meridian. The indices, m and n (n ranges from $-m$ to $+m$ in steps of 2), were the double index notation of the Zernike polynomials. Here, M indicated the order of Zernike polynomials. The Zernike coefficients, such as defocus, astigmatism, coma, primary spherical aberration (SA), were represented by the indices, a_n^m , in equation 1. For a hypothetical thickness distribution, let us assume a set of Zernike coefficients with magnitudes as follows: piston = $55 \mu\text{m}$; defocus = $-2 \mu\text{m}$; vertical coma = $-2 \mu\text{m}$; horizontal coma = $+2 \mu\text{m}$; SA = $1 \mu\text{m}$. Figure 1A shows a two-dimensional plot of a uniform thickness epithelium equal to the magnitude of piston. Figures 1B–E shows the thickness profiles obtained by multiplying the set of Zernike coefficients with their respective polynomials. By adding figures 1A–E a classical epithelium thickness profile (figure 1F) was obtained which was visually very similar to the epithelium thickness distribution observed in KC corneas. Thus, the Zernike decomposition was applied to both distributions of epithelium and total corneal thicknesses (epithelium plus stroma) and this captured the complete 3D spatial variation of thicknesses. In addition to the individual Zernike terms, the root mean squares of the thickness lower and higher order Zernike coefficients were also calculated (table 1).

- **Volume of epithelium and total cornea:** The volume was measured using the following integral equation 12:

$$V = \int_{-\pi}^{\pi} \int_0^R (t \times r) dr d\theta \quad (2)$$

Here, V was the volume, R was the analyses radius, t was the local thickness at a given radius (r) and meridian (θ). The total cornea included both stroma and epithelium. The ratio

of the volume of superior and inferior half of the epithelium and corneal 3D structure were also evaluated.

In the universal architecture, we used only the above indices since all the curvatures, aberrations, thicknesses and volumes were mapped in 3D by Zernike polynomials only. The order of the Zernike polynomials was 6 and the analyses diameter was fixed at 6.4 mm.^{3–8} The analysed diameter was chosen such that none of the eyes had missing elevation data due to incomplete edge detection¹¹ or lack of Placido coverage. Beyond 6.4 mm diameter, many disease eyes had missing epithelium thickness data due to inability of the scanner to detect the epithelium-BW interface accurately. Table 1 lists the various indices generated for each cornea. In a recent study, the random forest (RF) model achieved the best area under curve (AUC), accuracy and precision for detection of subclinical KC among all tested AI models.⁵ Our earlier studies on 3D Zernike analyses of epithelium and BW curvature also used the RF AI model.^{3–8} Therefore, we used the same RF AI in this study as well along with a fivefold cross-validation. The fold cross-validation is a standard method recommended for small data sets.¹³ In this, k unique folds (or parts) of the data were created. Then, $k-1$ parts were used to train and the one remaining part was used to test the model. There was no overlap of eyes between the test and train data, and this was automatically controlled by the software. Each RF model was limited to 10 decision trees and not less than 10 eyes per node. The AI analyses was performed using the Orange data mining package v3.26 (University of Ljubljana, Slovenia). It should be noted here that MS-39 did not have any in-built AI which combined curvature, thickness and aberrations data into one model.

Statistical analyses

The RF AI model was evaluated using the following indices: AUC, classification accuracy, precision, F1-score and log-loss. Ideally, a prediction completely matching the classification would yield an AUC of 1.0 and all percentages will tend to 1.0 (or 100%).

Table 2 Summary of classification achieved by the AI model

	Area under the curve	Classification accuracy	F1-score	Precision	Recall
Healthy	0.994	0.956	0.955	0.927	0.985
VAE	0.976	0.955	0.820	0.912	0.715
KC	0.997	0.991	0.989	0.991	0.987

AI, artificial intelligence; KC, keratoconus; VAE, very asymmetric ectasia.

The corresponding magnitude of log-loss would tend to zero. The mean±SD of a few of the top variables identified by the AI for best classification were also evaluated with the Kruskal-Wallis post hoc test (MedCalc, Ostend, Belgium). The sample size adequacy was also tested using the magnitudes of AUC (MedCalc). A $p<0.05$ was considered statistically significant.

RESULTS

Table 2 shows the performance metrics of the achieved classification. The AI achieved an AUC greater than 0.97 ($p<0.05$) for all groups of eyes. The F1-score, precision and recall in the diagnosis of healthy and KC eyes were comparable. However, the same (82%, 91.2% and 71.5%, respectively) were lower in VAE eyes, respectively. This was because a significant proportion of the VAE eyes had tomography features similar to the healthy eyes ($n=36$) and were reclassified by the AI accordingly. The classification accuracy was the highest for the KC eyes (**table 2**). The other AI performance measures indicated the same inferences (**table 2**). Based on the AUC of healthy eyes, the required sample size for positive group (healthy eyes in this case) was 6 and for negative group (VAE and KC eyes) was 7. For VAE eyes, the required sample size for positive group (VAE eyes in this case) was 5 and for negative group (healthy and KC eyes) was 34. For KC eyes, the same were 8 and 8, respectively. Thus, the sample sizes were adequate for this study even after fivefold cross-validation.

Table 3 shows commonly used corneal tomography variables based on clinical information. Except for AntLORMS, PostK2,

Table 3 Corneal variables (mean±SD) of the groups based on clinical information

	Healthy (1)	VAE (2)	KC (3)	P<0.001
AntK1 (D)	45.082±1.350	45.465±1.672	54.966±6.792	All*
AntK2 (D)	43.719±1.219	43.496±1.383	47.839±5.220	All
AntKmax (D)	45.312±1.356	45.753±1.750	56.249±7.515	All
AntLORMS (μm)	1.505±0.532	1.514±0.712	6.152±5.589	(3) vs (1), (2)†
AntHORMS (μm)	0.571±0.151	0.777±0.339	3.211±1.891	All
PostK1 (D)	-6.021±0.192	-5.898±0.377	-5.786±0.782	All
PostK2 (D)	-6.349±0.221	-6.364±0.345	-7.51±1.025	(3) vs (1), (2)
PostKmax (D)	-5.84±0.211	-5.672±0.384	-5.185±0.600	All
3D-LORMS (μm)	1.91±0.70	1.949±0.853	5.293±4.792	(3) vs (1), (2)
3D-HORMS (μm)	0.556±0.141	0.722±0.309	2.779±1.713	All
Min CT (μm)	541.32±28.97	500.55±38.24	443.14±40.30	All
Min ET (μm)	46.58±4.20	46.17±3.78	42.52±4.89	(3) vs (1), (2)
CCT (μm)	545.05±29.44	510.16±27.43	464.22±38.33	All
CET (μm)	54.75±3.64	52.87±4.19	49.06±6.86	All

The VAE eyes were the fellow eyes of KC patients. The last column shows statistical significance between groups.

*Indicates that all groups were different significantly from each other.

†Indicates that (3) differed significantly from both (1) and (2) while (1) and (2) were similar. CCT, central corneal thickness; CET, central epithelium thickness; HORMS, root mean square of higher order Zernike terms; KC, keratoconus; LORMS, root mean square of lower order Zernike terms; VAE, very asymmetric ectasia.

Table 4 Commonly used corneal parameters of the groups after AI training and cross-validation

	Healthy (1)	VAE-H (2)	VAE (3)	KC (4)
AntK1 (D)	45.071±1.339	45.122±1.539	45.471±1.447	55.057±6.776
AntK2 (D)	43.72±1.214	43.65±1.510	43.427±1.344	47.871±5.24
AntKmax (D)	45.296±1.335	45.315±1.590	45.797±1.541	56.353±4.792
AntLORMS (μm)	1.503±0.530	1.328±0.412	1.554±0.757	6.195±5.605
AntHORMS (μm)	0.567±0.144	0.611±0.187	0.816±0.309	3.237±1.885
PostK1 (D)	-6.023±0.192	-5.996±0.208	-5.852±0.422	-5.785±0.787
PostK2 (D)	-6.348±0.221	-6.302±0.259	-6.364±0.333	-7.523±1.022
PostKmax (D)	-5.841±0.209	-5.832±0.223	-5.618±0.414	-5.178±0.599
3D-LORMS (μm)	1.90±0.684	1.822±0.696	1.936±0.792	5.328±4.807
3D-HORMS (μm)	0.552±0.135	0.591±0.155	0.749±0.289	2.802±1.709
Min CT (μm)	541.69±28.97	520.23±20.91	493.47±40.79	442.19±39.50
Min ET (μm)	46.67±4.05	46.38±4.11	46.14±3.67	42.46±4.88
CCT (μm)	545.38±29.46	524.63±21.13	505.29±27.12	463.37±37.68
CET (μm)	54.75±3.63	52.35±3.17	53.20±4.50	48.98±6.85

The VAE and VAE-H eyes were the fellow eyes of KC patients. The last column shows statistical significance between groups.

AI, artificial intelligence; CCT, central corneal thickness; CET, central epithelium thickness; HORMS, root mean square of higher order Zernike terms; LORMS, root mean square of lower order Zernike terms; VAE, very asymmetric ectasia; VAE-H, VAE eyes re-classified as healthy.

3D-LORMS and min CET (or central epithelium thickness), all variables differed significantly between the healthy, VAE and KC eyes ($p<0.001$). For AntLORMS, PostK2, 3D-LORMS and min CET, only the KC eyes differed significantly from the healthy and VAE eyes. **Table 4** shows the same commonly used variables after training and cross-validation of the RF model. The AntK1 of healthy and VAE eyes reclassified as healthy (VAE-H) corneas differed significantly from VAE and KC corneas ($p<0.001$). The AntK1 of VAE corneas also differed significantly from KC corneas ($p<0.001$). A similar trend was observed for AntK2. The AntKmax, AntHORMS, PostKmax and 3D-HORMS of healthy and VAE-H corneas differed significantly from VAE and KC corneas ($p<0.001$). The AntLORMS, PostK2, 3D-LORMS and min ET of KC corneas differed significantly from the other groups ($p<0.001$). The central corneal thickness and min CT differed significantly among all the groups ($p<0.001$). The CET of healthy corneas differed significantly from the other groups ($p<0.001$). However, the CET of VAE-H and VAE were similar ($p>0.05$). The PostK1 of healthy and VAE-H corneas differed significantly from KC corneas ($p<0.001$). Thus, four parameters (AntKmax, AntHORMS, PostKmax and 3D-HORMS) in **table 4** indicated significant differences between VAE-H and VAE corneas while indicating similar magnitudes of the same parameters between healthy and VAE-H corneas. Since a significant proportion of the VAE-H ($n=36$) were classified as healthy, we evaluated a few of the top classification variables based on information gain of the RF AI model. These variables are summarised in **table 5** in the order of highest to lowest rank based on information gain.

The root mean square of BW coma wavefront aberration (BWComaRMS) was 0.251 and 0.365 μm for the healthy and VAE-H ($p>0.05$). The same was 0.622 and 3.242 μm for most of the remaining VAE ($n=104$) and KC eyes, respectively ($p<0.001$). Overall (**table 5**), most variables indicated that the healthy and VAE-H ($n=36$) eyes differed significantly from most of the remaining VAE ($n=104$) and KC eyes ($p<0.001$). Interestingly for all the variables (**table 5**), the VAE-H group differed

Table 5 Summary of few of the top variables used primarily by the AI for classification of eyes

	Healthy	VAE-H*	VAE†	KC†
BWComaRMS (μm)	0.251 $\pm$ 0.005	0.365 $\pm$ 0.013	0.622 $\pm$ 0.035	3.242 $\pm$ 0.082
BWHORMS (μm)	0.534 $\pm$ 0.006	0.587 $\pm$ 0.029	0.967 $\pm$ 0.046	4.02 $\pm$ 0.092
AntHORMS (μm)	0.564 $\pm$ 0.006	0.613 $\pm$ 0.030	0.80 $\pm$ 0.030	3.243 $\pm$ 0.089
AntComaRMS (μm)	0.25 $\pm$ 0.005	0.341 $\pm$ 0.018	0.48 $\pm$ 0.026	2.681 $\pm$ 0.077
tHORMS (μm)	1.585 $\pm$ 0.026	1.516 $\pm$ 0.080	2.726 $\pm$ 0.214	8.81 $\pm$ 0.200
tComaRMS (μm)	0.736 $\pm$ 0.021	0.667 $\pm$ 0.064	1.557 $\pm$ 0.146	6.535 $\pm$ 0.174
BWKmax (Diopter)	45.805 $\pm$ 0.061	45.932 $\pm$ 0.312	46.817 $\pm$ 0.169	59.194 $\pm$ 0.368
AntKmax (Diopter)	45.289 $\pm$ 0.059	45.199 $\pm$ 0.282	45.783 $\pm$ 0.146	56.375 $\pm$ 0.353
BWK1 (Diopter)	45.333 $\pm$ 0.061	45.422 $\pm$ 0.303	46.211 $\pm$ 0.148	57.318 $\pm$ 0.326
eComaRMS (μm)	0.376 $\pm$ 0.013	0.291 $\pm$ 0.037	0.622 $\pm$ 0.044	1.965 $\pm$ 0.050
eHORMS (μm)	0.873 $\pm$ 0.015	0.868 $\pm$ 0.047	1.29 $\pm$ 0.071	3.059 $\pm$ 0.059
tHORMS (μm)	1.585 $\pm$ 0.026	1.516 $\pm$ 0.080	2.726 $\pm$ 0.214	8.81 $\pm$ 0.200
Min CT (μm)	541.6 $\pm$ 28.9	526.7 $\pm$ 22.3	495.5 $\pm$ 41.3	442.1 $\pm$ 39.5
Min ET (μm)	46.6 $\pm$ 4.1	47.1 $\pm$ 3.9	45.9 $\pm$ 3.7	42.5 $\pm$ 4.9

For definition of variables, refer to table 1. The last two rows present the minimum corneal and epithelium thicknesses of the respective groups. The variables were listed in the order of highest to lowest rank based on information gain.

*The variables of VAE-H eyes were significantly different from healthy eyes ($p<0.001$).

†The variables were significantly different from healthy and VAE-H eyes ($p<0.001$).

AI, artificial intelligence; BW, Bowman's layer; ComaRMS, root mean square of coma Zernike terms; HORMS, root mean square of higher order Zernike terms; KC, keratoconus; VAE-H, very asymmetric ectasia eyes reclassified as healthy.

significantly from healthy eyes ($p<0.001$) despite the AI identifying these eyes as healthy. Further, the variables summarised in table 5 showed that both epithelium and corneal thickness Zernike variables were highly indicative of the features of the study groups instead of central and minimum values of the thicknesses. These inferences were true for most of the other variables summarised in table 5. Figure 2 shows the bar plots of mean and SE of the mean for some of the variables listed in table 5.

We also evaluated the novel analyses of epithelium thickness distributions with Zernike polynomials between the groups. Figure 3A shows the dot plots of central (open circles) and minimum (filled squares) epithelium thickness of the respective groups as classified by the AI. Similarly, figure 3B shows the eLORMS and eHORMS (table 1) of the respective groups as classified by the AI. The dot plots clearly showed the overlap of the magnitude of thicknesses (figure 3A) and thickness distributions analysed by Zernike polynomials (figure 3B) between the different groups despite significant statistical differences between the groups (eHORMS and tHORMS in table 5). For example, the tHORMS of healthy and VAE-H eyes differed significantly from most of the remaining VAE and KC eyes ($p<0.001$). Figure 3C shows the results of the regressions between eLORMS and eHORMS (table 1) with minimum ET (table 1) for all the eyes irrespective of classification. It was evident that the distortion in thickness distribution increased with increasing severity of the KC, that is, slope of the regression lines (figure 3C) was less than zero ($p<0.001$). Intuitively, greater the spatial asymmetry (distortion) of thickness distribution indicated by eLORMS and eHORMS, the greater would be the severity of KC. figure 4 shows an example of a VAE-H by the AI. Figure 5 shows an example of a VAE eye retained as a VAE by the AI.

DISCUSSION

There are no universal criteria to define subclinical and FFKC.⁹ Some studies used cut-off's for commonly used variables to

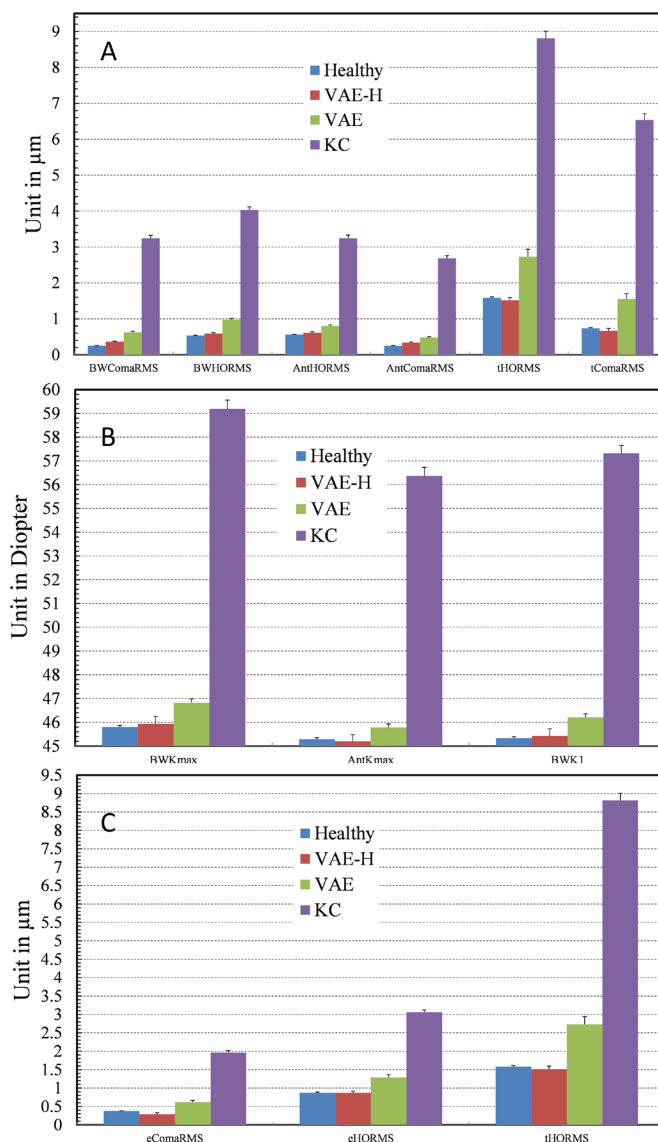


Figure 2 (A–C) Bar plots of few of the top variables (shown in table 5) identified by the random forest AI model. AI, artificial intelligence; eComaRM, root mean square of coma Zernike terms; eHORMS, root mean square of higher order Zernike terms; KC, keratoconus; VAE-H, very asymmetric ectasia eyes reclassified as healthy.

define subclinical KC and FFKC, for example, a central keratometry (K) greater than 47.2 diopters (D) and an inferior–superior (I–S) value greater than 1.4 D were used.¹⁴ These may lead to an inaccurate classification of eyes and comparison of AI/DL models from different studies. Another aspect which is often ignored in studies is the quality of the edge detection in tomography images. While it is not possible to assess proprietary edge detection algorithms used by different tomographers, a visual examination of missing edges in the cross-sectional images should be conducted before any AI/DL analyses.¹¹ In this study, we classified the fellow asymmetric eye as VAE only and did not perform manual sub-classification of these eyes further into subclinical KC and FFKC. This approach was similar to the one used in a recent study.¹⁵ In the earlier study, all VAE eyes were considered as subclinical KC and this group also included eyes which were termed as FFKC according to the definitions of Klyce.^{15 16} In contrast, we chose to assign the fellow asymmetric eyes of KC

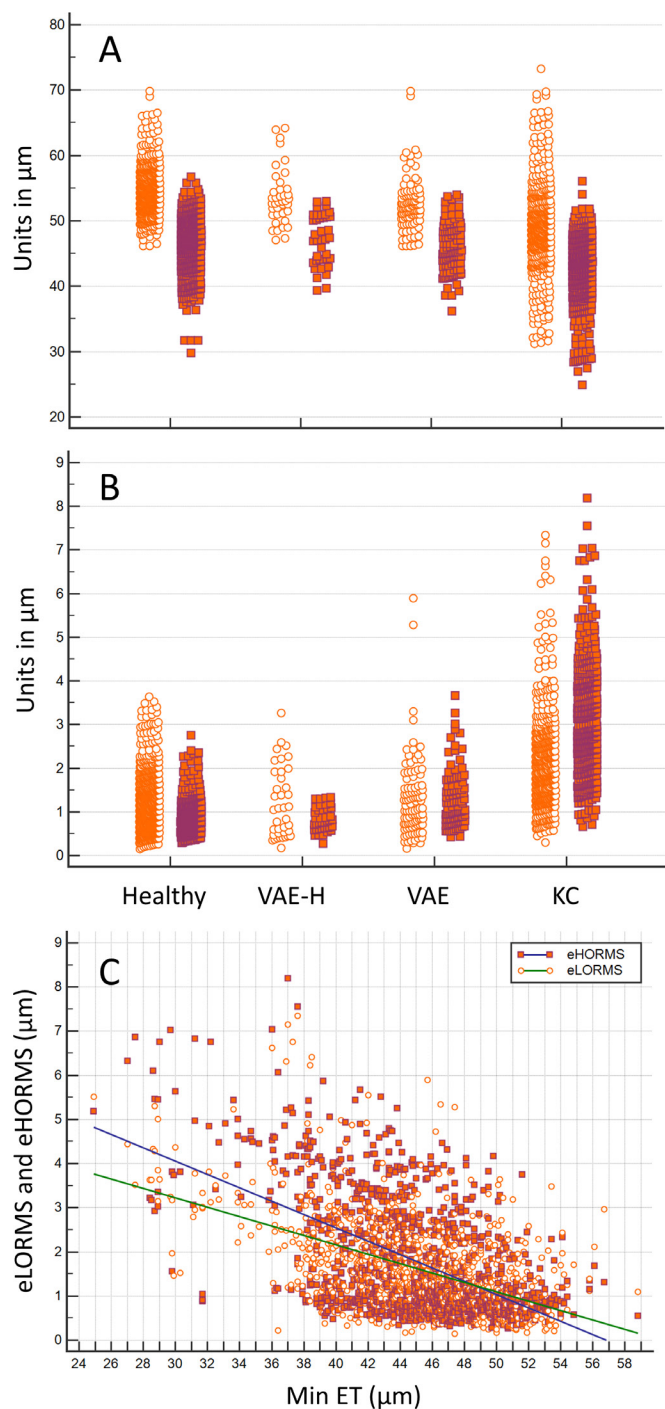


Figure 3 (A) Dot plot of central (open circles) and minimum (filled squares) epithelium thickness of the respective groups as classified by the AI; (B) dot plot of eLORMS (open circles) and eHORMS (filled squares) of the respective groups as classified by the AI; (C) regression of eLORMS and eHORMS with the minimum epithelium thickness (min et). AI, artificial intelligence; eHORMS, root mean square of higher order Zernike terms; eLORMS, root mean square of lower order Zernike terms; KC, keratoconus; VAE-H, very asymmetric ectasia eyes re-classified as healthy.

patients simply as VAE for training and cross-validating the RF model.

Interestingly, the universal architecture and AI model achieved precisely the intended segregation of the VAE eyes since the VAE-H eyes ($n=36$) could be termed as ‘subclinical’ and the

remaining VAE ($n=104$) eyes could be termed as ‘FFKC’. Intuitively, this was reasonable to conclude since the VAE-H eyes ($n=36$) had curvature, wavefront aberrations and thickness distributions somewhat different from the healthy eyes (table 5). Table 4 also shows that some of the variables of VAE-H eyes were similar to the healthy eyes while the rest showed difference between the healthy and VAE-H eyes. Table 5 indicates that all the top-ranking variables differed significantly between the VAE-H and healthy eyes. Hence, these variables (table 5) were primarily used by the AI for reclassification of the 36 eyes. Further, all the variables in table 4 differed significantly between the healthy and VAE eyes ($p<0.001$). Further, most of the remaining VAE eyes had features which preceded the features of KC eyes but were still distinctly different from healthy and VAE-H eyes. Figure 2 and table 5 supported the above conclusions. This outcome of the AI training and cross-validation supported the comments discussed in a recent review,⁹ which stated that treating subclinical condition as diseases evident on imaging but not on clinical examination was an incorrect classification and FFKC definition should follow the original definition by Amsler. Figures 4 and 5 provide evidence as early disease changes may be seen in the latter. Since the model effectively identified the VAE-H and VAE eyes as ‘subclinical’ and ‘FFKC’, it was clear that any new model based on the four groups identified by our RF AI would provide a perfect AUC, CA, precision, sensitivity and specificity of ~ 1.0 or 100% for prospective classification of corneas as healthy, VAE-H, VAE and KC. Thus, the use of our universal architecture and four groups identified by the AI would eliminate the uncertainties of visual diagnosis of corneal topography among clinicians. For example, the accuracy of classification was 92.8% among senior refractive surgeons but only 85.6% among student refractive surgeons and 68.2% among senior non-refractive surgeons in a study and yet the DL model performed better (95%) than all of them.¹⁷ In this study, the RF AI had no other option but to re-classify some ($n=36$) of the VAE eyes as healthy since it did not have a four-group classification for training and cross-validation to begin with. The results from table 5 supported the above inference since the variables of VAE-H eyes were significantly different from healthy eyes ($p<0.001$). Thus, a VAE eye of an asymmetric KC patient would most likely be assigned into either healthy or ‘true’ VAE group by our RF model. In the event, a VAE eye is assigned as healthy by the RF model, then this eye will be considered as a VAE-H eye. The range of data indicated by means and standard deviation (table 5) will drive the performance of our RF model.

In our recent study, we performed manual classification of anterior corneal surface and BW variables into healthy, VAE, FFKC and KC eyes.³ Interestingly, most of the VAE and some of the FFKC eyes with normal features were re-classified as healthy by the AI.³ This showed that AI models were capable of identifying VAE eyes which were ‘subclinical’ and yet to progress to FFKC or frank VAE on tomography. In a recent study, the anterior and posterior surface variables along with demographics of the patients achieved an AUC of 0.96, precision of 89%, accuracy of 87%, sensitivity of 88% and specificity of 85% using an RF model for classification of subclinical eyes.⁵ However, the study defined subclinical eyes as eyes with abnormal topography, I-S localised steepening or asymmetric bowtie pattern.⁵ Another study using Scheimpflug tomography imaging considered both FFKC and subclinical KC to be the same.¹⁸ Such a definition of subclinical KC was incorrect since these eyes were still abnormal on imaging and should have been termed as FFKC instead.⁹ Another study performed separate bivariate comparisons of healthy versus FFKC and healthy versus KC and used device

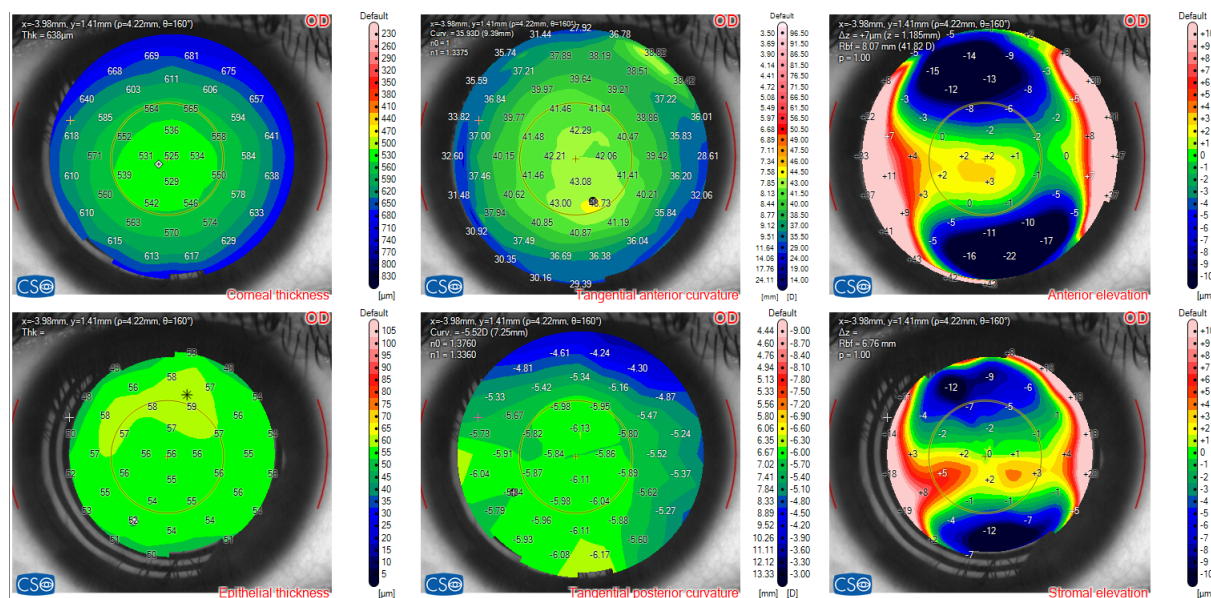


Figure 4 MS-39 imaging plots of a VAE eye reclassified as healthy (VAE-H or subclinical) by the AI model. AI, artificial intelligence; VAE-H, very asymmetric ectasia eyes reclassified as healthy.

in-built AI model of curvature (anterior and posterior) and local measurements of corneal thicknesses, for example, as central thickness, to segregate the eyes.¹⁰ The study concluded that very mild forms of ectasia may be undetectable by the in-built AI model of the device.¹⁰ We speculate that the lack of a 3D quantification of segmental tomography may be the reason for the outcome of the earlier study.¹⁰ Another study used Pentacam (Scheimpflug imaging) device parameters only, and achieved an AUC, sensitivity and specificity of 0.96, 91% and 90%, respectively, in the diagnosis of preclinical KC.¹⁴ Here as well, the study had defined cut-off's of variables for identification of preclinical KC.¹⁴ It should be noted that using defined cut-off's of variables for diagnosis inherently introduces a bias in the resulting AI/DL models.

Since different studies used different variables and these variables may not be common to all tomographers, the comparison of outcomes of different studies becomes challenging. Nonetheless, our results (table 2) were better than the results from other studies, and also were based on an AI driven classification of subclinical (VAE-H) and FFKC (VAE) using segmental tomography and 3D analyses of the cornea. This was also highlighted in a recent study where Scheimpflug (anterior and posterior) and OCT (sectoral epithelium thicknesses) variables were combined for an effective classification of healthy and highly asymmetric KC.¹⁹ However, the study used very small sample sizes, relied on device dependent parameters and surgeon defined criteria of asymmetric eyes prior to the training of logistic regression model.¹⁹ In another study, all fellow asymmetric eyes were

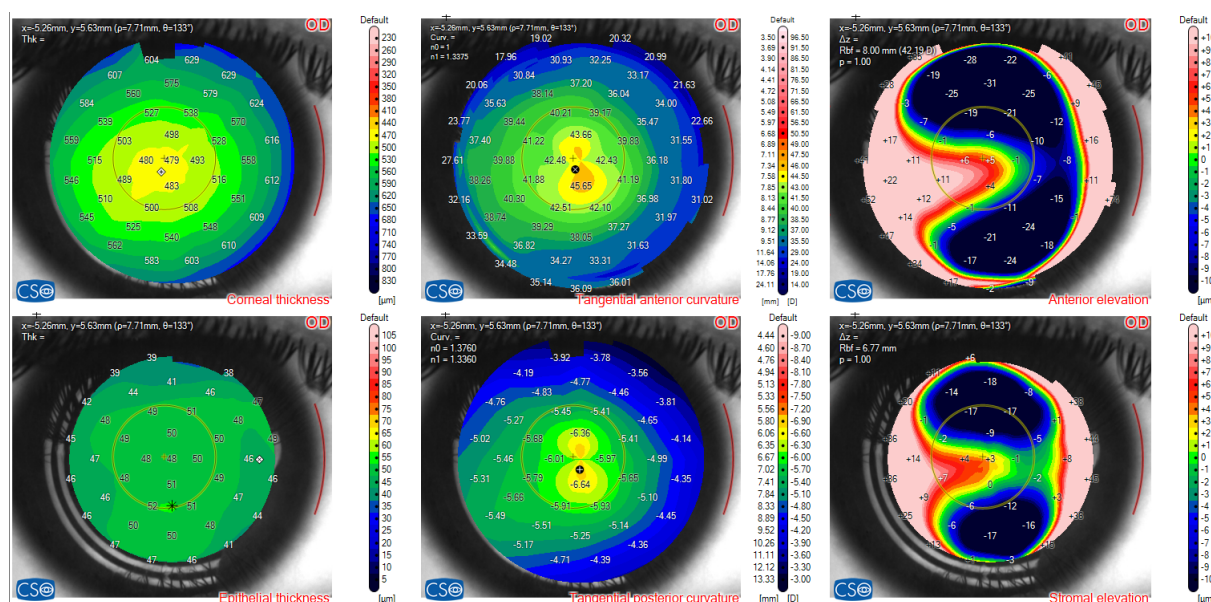


Figure 5 MS-39 imaging plots of a VAE eye retained as VAE (forme fruste) by the AI model. AI, artificial intelligence; VAE-H, very asymmetric ectasia eyes reclassified as healthy.

classified as FFKC for the training of a support vector machine AI classifier and a significant number of FFKC eyes were reclassified as healthy by the AI.²⁰ In comparison to our study (table 2), the earlier study achieved an AUC, sensitivity, specificity, precision and accuracy of 0.925, 37.3, 98%, 52% and 88%, respectively.²⁰ A combination of curvature, corneal thickness and anterior elevation was the best discriminant between healthy and KC eyes, and a combination of curvature, thickness and posterior elevation was the best discriminant between healthy and subclinical KC eyes.²¹ Our universal architecture of AI and analyses with Zernike polynomials for 3D corneal assessment precisely advanced the above findings.²¹

In our study, we presented and validated a universal architecture of analysing segmental tomography which combined Placido and OCT data. Here, the Zernike analyses was very powerful since it decomposed the complete 3D distribution of variables using well known descriptors such as defocus, coma, primary SA. Intuitively, it was expected that local measurements of curvature and thicknesses would correlate significantly with Zernike variables and an example of this was shown in figure 3C. The Zernike polynomials are already available in the software of commercially available tomographers. Therefore, it would be relatively easy to implement this universal architecture with the aid of RF in existing (MS-39) and future devices. There are limitations in this study though. We chose not to include surgeon derived classification of subclinical and FFKC for the training of the AI model. There may be disagreement between surgeons regarding the assessment of an individual cornea. Nonetheless, our machine-driven approach was precisely intended to avoid these disagreements in future diagnosis of early disease. The true assessment of whether a subclinical or FFKC progresses to clinical KC can occur only with longitudinal assessment. Since this assessment was lacking in this study, we intend to perform longitudinal studies in future using this same universal architecture.

Another limitation was that no biomechanical assessments were combined into the universal architecture. Using the Corvis-ST (OCULUS Optikgeräte, Germany), the Total Biomechanical Index (TBI) achieved the best AUC of 0.79, sensitivity of 67% and specificity of 86% in diagnosis of subclinical KC from healthy eyes.²² Here, the subclinical KC eyes were isolated based on predefined cut-off's of mean keratometry less than 47.2D, I-S asymmetry less than 1.4D and KC percentage index less than 60%.²² In another study, 12 patients had VAE in one eye and the other eye was *frank* KC.²³ Here, the Corvis Biomechanical Index was abnormal in both the eyes.²³ In another study, 37 healthy and 23 subclinical KC eyes were compared with Corvis-ST variables.²⁴ While all tomographic variables differed between the two groups, only 2 Corvis-ST variables differed between the healthy and subclinical eyes and the TBI achieved an AUC of 0.925 (sensitivity=84.4% and specificity=82.4%).²⁴ Here as well, tomography was used for the diagnosis of subclinical KC eyes before biomechanical variables were compared.²⁴ In another study, subtle tomographic changes were noted in subclinical eyes (VAE fellow eye) despite a healthy biomechanical response measured by the Corvis-ST compared with the KC eye.²⁵ Nearly 40% of the VAE eyes with apparent normal tomography were classified as healthy by the Corvis-ST variables and VAE was established based on predefined cut-off's of topographic indices such as the Klyce/Maeda Keratoconus Index.²⁶ In all these studies, tomography was limited to the anterior and posterior corneal surface and detailed 3D tomographic comparisons of segmental tomography was lacking. Thus, there is some evidence to support that the biomechanical changes may occur in early ectasia but further studies are required with the Corvis-ST and preclinical technologies.²⁶ We may speculate that the VAE-H

eyes (n=36) in our current study probably had a healthy biomechanical profile similar to the healthy eyes. In summary, a novel universal architecture using Zernike polynomials analyses of curvature, wavefront aberrations and segmental thicknesses (epithelium and cornea) and AI was presented. The AI model effectively classified the VAE eyes as eyes with healthy tomography (VAE-H) and early disease (VAE). Further studies with different regional populations and external validation is required to confirm the robustness of the model. Alternate mathematical formulations of Zernike polynomials, which may provide better analyses of corneal tomography, need to be evaluated in future studies using this universal architecture.^{27 28}

Contributors GK, RS, PK, RM and SG were involved in data collection. GK, PK, RM, SG and ASR were involved in data analyses. GK, RS, ASR and RM were involved in writing of the manuscript. RS, ASR and RN were involved in critical revision of the manuscript. RS, PK and ASR were involved in design of the study.

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Competing interests RS and ASR have a USA patent on imaging of Bowman's layer and its application for assessing progression of disease.

Patient consent for publication Not applicable.

Ethics approval The study was approved by the ethics committee of the Narayana Nethralaya eye hospital, India. The study followed the tenets of the Declaration of Helsinki.

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